HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented API-level safeguards including **rate-limiting** and **parameter validation** for a high-volume transaction processing system, reducing incident rates by **18%**.
- Translated attacker telemetry and threat intelligence into precise technical abuse requirements, defining control specifications for cross-cutting vectors impacting payment and onboarding surfaces.
- Developed automated regression suites and attack simulations in Python to ensure mitigated abuse vectors do not recur, achieving **95%** test coverage for critical control logic.
- Collaborated with Abuse Research and Security teams to co-design resilient controls across payment, onboarding, and identity surfaces, balancing risk mitigation with user conversion.
- Prototyped and deployed technical controls across API and protocol boundaries to address high-impact abuse vectors, contributing to the protection of a financial ecosystem processing millions of transactions.
- Executed control incubation by defining handoff criteria and operational documentation, successfully transferring mature defenses to product teams with minimal operational impact.
- Containerised backend services using **Docker** and deployed to an **AWS EKS** cluster, implementing basic resource limits and health checks for improved stability.

SKILLS

Languages – Python, Java, SQL, Go

Cloud & Containerization – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes (EKS)

Frameworks & Libraries – Spring Boot, FastAPI, Flask, pandas, NumPy, scikit-learn

Databases – PostgreSQL, MySQL, Oracle

Tools & Methodologies – Jenkins, Git, Jira, Confluence, Agile, Scrum, Splunk, Dynatrace, A/B Testing

PROJECTS

API Abuse Control Implementation (HCL Technologies)

- Developed Python scripts to analyze API access logs from enterprise customers, identifying patterns indicative of automated abuse and brute-force attacks.
- Implemented **rate-limiting** mechanisms and **input validation** checks within a Java Spring Boot application to protect against common API abuse vectors, reducing suspicious request volume by **22%**.
- Wrote comprehensive unit and integration tests for new control logic using JUnit, ensuring adherence to security specifications and functional correctness.

Automated Threat Regression Suite (MSc Applied Project)

- Designed and built an automated testing framework in Python to simulate common attacker TTPs against simulated financial transaction endpoints.
- Integrated the framework with **scikit-learn** for anomaly detection on simulated user behaviour, achieving an AUC of **0.81** in identifying bot-like activity.
- Documented the testing methodology and results, providing actionable insights for enhancing system resilience against evolving threats.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)